

SHEIKH HEFZUL BARI

Email: shbari.bd@gmail.com

Education

PhD Symbiotic Environment Systems, Fukushima University, Japan (Expected date: March 2026)

Thesis title: 'Material Transport from Watershed to Ocean'

Supervisor: Dr. Yoshiyuki Yokoo

Synopsis: A combination of literature review, experimental research, and statistical analysis of suspended sediment dynamics in a torrential watershed.

MSc. Civil & Environmental Engineering, Shahjalal Uni. of Science & Technology (2014)

BSc. Civil & Environmental Engineering, Shahjalal Uni. of Science & Technology (2011)

Scholarships and Awards

Japanese Government (MEXT) Scholarship – PhD Fellowship

National Science and Technology (NST) Fellowship – MSc., Ministry of Science and Technology, Bangladesh

Best Researcher Award 2017– Leading University, Bangladesh

Merit-Based Scholarships – Ministry of Education, Bangladesh (Secondary & Higher Secondary Scholarships)

Technical Skills

Programming **Python** user for scientific plotting, data processing, and analysis of small to medium datasets.

Languages Native in **Bengali**, communicative in **English** with the capacity for scientific writing, and basic proficiency in **Japanese** and **Hindi** for practical communication.

Research Grants

Estimating probable maximum precipitation for Bangladesh (As PI)

Leading University Research Grant (Partially Funded). Time: January 2019 - December 2019

Assessing climate change impact in Haor Basin of Bangladesh (As PI)

Funded by Leading University Research Grant. Time: January 2017 - December 2017

Development of low-cost desalination device (As PI)

Ministry of Science and Technology, Bangladesh. Time: January 2015 - March 2016

Research and Academic Experiences

Research Assistant Watershed Hydrology lab, Fukushima University, April, 2023 – March, 2025

Faculty Member (On study Leave) Department of Civil Engineering, Leading University, Sylhet, Bangladesh, August 2013 – Present

Research Assistant Department of Civil & Environmental Engineering, Shahjalal University of Science and Technology, Sylhet, Bangladesh, January 2012 – July 2013

Conference Presentations and Lectures / Invited Talk

*“A reconnaissance study on estuarine suspended sediment dynamics in the Abukuma River, Japan.” Japan Geoscience Union (JpGU) Meeting 2025.

“Preliminary field study for understanding estuarine suspended sediment dynamics at the Abukuma River, Japan.” Japan Society of Civil Engineers Annual Meeting 2024, Sendai. February 28-March 01 2025.

Publications *(Selected journals are listed here only)*

Bari, SH., Yokoo, Y., Tateno, M. and Leong, C. (2025). Dynamics, particle size distribution and mineral composition of suspended sediment transport in the Abukuma River estuary. *Hydrological Research Letters*, 19(1), pp.51-57.

Bari, SH., Yokoo, Y. and Leong, C. (2024). A brief review of recent global trends in suspended sediment estimation studies. *Hydrological Research Letters*, 18(2), pp.51-57.

Husna, NEA., **Bari, SH.**, Islam, GT. and Islam, AS. (2024). Groundwater drought assessment in Southwestern Bangladesh. *Int. J. of Hydrology Science and Technology*, 17(3), pp.267-277.

Husna, NEA., **Bari, SH.**, Islam, GT., Islam, AS. and Hussain, MM. (2024). Modelling drought as a climate change indicator in the southwest coastal region of Bangladesh. *Int. J. of Hydrology Science and Technology*, 18(4), pp.351-67.

Hussain, MM., Mahmud, I. and **Bari, SH.** (2023). pyHomogeneity: a python package for homogeneity test of time series data. *J. of Open Research Software*, 11(1).

Bari, SH. and Husna, NEA. (2023). People's perception on climate change effects and adaptation in the Haor basin of Bangladesh. *Int. J. Hydrology Science and Technology*, 15(1), pp.40-53.

Bari SH. (2021). Estimating probable maximum precipitation for Bangladesh. *Int. J. Hydrology Science and Technology*, 11(4), pp.415-421.

Bari, SH., Hussain, MM. and Husna, NEA. (2017). Rainfall variability and seasonality in northern Bangladesh. *Theoretical and Applied Climatology*, 129(3-4), pp.995-1001.

Mahmud, I., **Bari, SH.** and Rahman, MT. (2017). Monthly rainfall forecast of Bangladesh using autoregressive integrated moving average method. *Environmental Engineering Research*, 22(2), pp.162-168.

Bari, SH., Hoque, MA., Rahman, MT. and Hussain, MM. (2016). Analysis of seasonal and annual rainfall trends in the northern region of Bangladesh. *Atmospheric Research*, 176-177, pp.148-158.

Husna, NEA., **Bari, SH.**, Hussain, MM., Rahman, MT. and Rahman, M. (2016). Ground water level prediction using artificial neural network. *Int. J. Hydrology Science and Technology*, 6(4), pp.371-381.

Hussain, MM., **Bari, SH.**, Tarif, ME., Rahman, MT. and Hoque, MA. (2016). Temporal and spatial variation of groundwater level in Mymensingh district, Bangladesh. *Int. J. Hydrology Science and Technology*, 6(2), 188-197.

Bari, SH., Rahman, MT., Hussain, MM. and Ray, S. (2015). Forecasting Monthly Precipitation in Sylhet City Using ARIMA Model. *Civil and Environmental Research*, 7(1), 69-77.

References

Dr. Yoshiyuki Yokoo | yokoo@sss.fukushima-u.ac.jp

Professor, Faculty of Symbiotic Systems Science, Fukushima University, Japan.

Dr. Tauhid-Ur-Rahman | tauhid_cee@yahoo.com

Professor, Dept. of Civil Engineering, Military Institute of Science and Technology, Bangladesh.